Pratyush Shendre

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Professional Summary

Software Development Engineer with a B.Tech. in AI/ML and hands-on experience building Python-based AI systems, computer-vision projects, and concurrent data workflows. Brings foundations in deep learning, machine learning, AI algorithms, software design, and data analysis to performance-focused AI engineering.

Experience

Software Development Engineer Jan. 2025 – Present
Amadeus Software Labs *India*

- Engineered and deployed a Meta-RAG framework for a Loyalty Portal chatbot, using multi-stage contextual validation to improve response grounding and reduce LLM hallucinations.
- Built data-driven reporting workflows that combine Splunk application telemetry with SQL data; used Python multiprocessing to execute independent queries concurrently with structured logging and progress reporting.
- Developed Python and Java automation for application-security and modernization workflows, including a Black Duck API v2 tool that reduced manual security-developer work by 40%.

AI–ML Virtual Intern May 2023 – Jul. 2023
AWS Academy *Virtual*

- Implemented end-to-end machine-learning pipelines with Amazon SageMaker, covering problem formulation, data collection, feature engineering, model training, deployment, and evaluation.

Technical Skills

AI / ML: Python, TensorFlow, Machine Learning, Deep Learning, Computer Vision, YOLOv8, OpenCV, CNNs, Natural Language Processing, RAG, Meta-RAG, Azure OpenAI
Programming / Data: Java, C, SQL, Python Multiprocessing, Software Design, Data Structures and Algorithms, Splunk, PostgreSQL, ChromaDB
Engineering: Git, Docker, Kubernetes, Linux, REST APIs, CI/CD, Azure, AWS, Jenkins

Selected Projects

GMS Knowledge Bot | ***Python, LangChain, Azure OpenAI, ChromaDB*** 2025

- Built an internal knowledge assistant that classifies questions by intent and adapts document retrieval and answer-generation settings for each query.
- Implemented retrieval and generation with LangChain, Azure OpenAI, and ChromaDB; created a Streamlit interface showing routing decisions, answers, logs, and session query history.

Crowd Detection & Management System | ***Python, YOLOv8, OpenCV, Computer Vision*** 2023

- Collected and labeled data, then built a crowd-density detection pipeline using YOLOv8 and computer-vision algorithms with a reported 80% accuracy.
- Implemented a real-time Telegram alert bot to support crowd-management response.

Red Light Violation Detection | ***Python, TensorFlow, YOLOv8, CNNs*** 2024

- Designed and implemented a computer-vision system to identify traffic-signal violations using deep-learning technologies.

Certifications & Achievements

- **Claude Certified Architect, Foundations** – Anthropic
- **GATE Qualified** (2023)
- Solved 1,000+ LeetCode problems; led planning and execution of technical events as a Technical Club Member at RCOEM Nagpur.

Education

Bachelor of Technology in Computer Science and Engineering (AI & ML) Aug. 2021 – Jul. 2025
RCOEM Nagpur *CGPA: 8.23/10*

- Coursework: Machine Learning, Deep Learning, Neural Networks, Data Structures and Algorithms (Java), Design and Analysis of Algorithms, Operating Systems, and Computer Networks.